Project Synopsis

1. Title - **Image Caption Generator Using Deep Learning**

Team Details

Student Name	College Name	Branch	Mobile No	Mail ID	Role (Lead/Member)
S. Pranavi	Malla Reddy Engineering College For Women	Data Science	9014925004	sandadipranavi2004@gmail.com	Lead
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Program Track and Domain

Program Track (Engg-HUB / Engg-Spoke / Science-HUB / Science-Spoke / Commerce-Spoke)	Domain (Analytics / Machine Learning / Deep Learning / GenAI)
Engg-Spoke	Deep Learning

Guide Details

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SAP

2. Problem Statement

With the rapid growth of digital images on the internet, manually describing images is time-consuming and inefficient. There is a need for an automated system that can:

- Understand the content of an image
- Identify objects and activities of a image
- Generate meaningful and grammatically correct captions

Existing traditional image processing methods cannot understand semantic meaning. Therefore, an intelligent Deep Learning-based system is required to generate accurate captions automatically.

3. Objectives

- To develop an automated image captioning system
- To implement a sequence generation model
- To integrate Computer Vision and NLP techniques
- To improve caption quality using optimization techniques

4. Methodology / Implementation Plan

1. Data Collection

- Download Flickr8K dataset
- Image folder (8,000 images)
- Caption file (40,000 captions)

2. Data Preprocessing

A. Image Preprocessing

- Resize images (e.g., 224×224 for CNN input)
- Normalize pixel values
- Convert images into arrays

B. Feature Extraction

- Use pretrained CNN model
- Remove final classification layer
- Extract high-level feature vectors

C. Text Preprocessing

- Convert captions to lowercase
- Remove punctuation and special characters
- Add start and end tokens (<start>, <end>)

Tokenization

- Build vocabulary
- Convert words to sequences
- Apply padding for equal length sequences

3. Model Design

A. Encoder (CNN)

- Input: Image
- Output: Feature vector

B. Decoder (LSTM)

- Input: Image feature + partial caption
- Embedding layer
- LSTM layer
- Dense layer with Softmax activation
- Output: Probability distribution of next word

4. Model Training

- Combine image features and text sequences
- Define loss function (Categorical Cross-Entropy)
- Use optimizer (Adam)
- Train model on training dataset
- Validate using validation dataset

- Save best model weights

5. Caption Generation Process

- Input new image
- Extract features using CNN
- Start with <start> token
- Predict next word
- Append predicted word
- Repeat until <end> token or max length

6. Model Evaluation

- Evaluate performance using:
- BLEU Score

5. Resources and Timeline

A. Hardware & Software Requirements

- Processor: Intel i3 / AMD equivalent
- RAM: 4 GB
- Storage: 10–20 GB free space
- System Type: 64-bit system
- GPU: NVIDIA GPU

B. **Dataset**
The dataset used for this project is publicly available.

We used the Flickr8K Dataset, which is widely used for image caption generation research.

Source link: https://www.kaggle.com/datasets/adityajn105/flickr8k

C. Project Timeline

Phase	Task	Tentative Completion Date
Phase 1	Problem Definition & Literature Review	20 February 2026
Phase 2	Dataset Collection (Flickr8K)	21 February 2026
Phase 3	Data Preprocessing (Image & Text Cleaning)	21 February 2026
Phase 4	Feature Extraction using CNN	23 February 2026

Phase	Task	Tentative Completion Date
Phase 5	Model Design (LSTM Decoder Architecture)	24 February 2026
Phase 6	Model Evaluation (BLEU Score)	24 February 2026
Phase 7	Caption Generation & Optimization (Beam Search)	25 February 2026
Phase 8	Documentation & Final Report Submission	26 February 2026

6. Expected Outcomes / Conclusion

- The Deep Learning-based Image Caption Generator successfully combines Computer Vision and NLP techniques to generate meaningful captions for images.
- The model trained on Flickr8K dataset can generate human-like descriptions with reasonable accuracy. This project demonstrates how CNN and LSTM networks work together to understand visual data and convert it into natural language.

7. References

- Flickr8K Dataset Repository.
- Deep Visual-Semantic Alignments for Generating Image Descriptions, CVPR
- Research publications on CNN–RNN based image captioning models.